

Research Statement - Christian Schmidt

Immediately after the Big Bang, the universe was comprised of hot, dense nuclear matter. This matter can be characterized as a “soup” of free quarks and gluons, or the elementary particles establishing the basic structure of all nuclei [1]. Experimentally, conditions similar to those of the early universe can be reconstructed via *relativistic heavy-ion collisions*, in which large nuclei collide at nearly the speed of light, forming microscopic droplets of the primordial *quark-gluon plasma* (QGP). However, a full *evolutionary* description of the universe can only be provided once an *equation of state* (EOS) is uncovered which encodes the relationships between its thermodynamic state variables.

Certain thermodynamic observables can be measured which provide insight into the EOS of nuclear matter. The observable of particular interest to my study is the squared speed of sound c_s^2 , which quantifies the stiffness of the medium, therefore giving direct insight to the thermodynamics of that medium. In a relativistic fluid, c_s^2 is given by the following:

$$c_s^2 = \frac{\partial P}{\partial \epsilon} = \frac{d \ln(T)}{d \ln(s)}, \quad (1)$$

where P and ϵ are the pressure and density of the medium, and T and s are the temperature and entropy density of the medium, respectively [1]. Recent hydrodynamic studies [2] argue that final state collision observables are sensitive to the thermodynamics of the initial state. More specifically, the mean transverse momentum $\langle p_T \rangle$ is linearly related to the initial state temperature of the medium, and the charged-particle multiplicity $\langle N_{ch} \rangle$ is directly analogous to the initial state entropy density. These proportionalities enable us to utilize eq. (1), motivating the extraction of c_s^2 from a power-law:

$$c_s^2 = \frac{d \ln \langle p_T \rangle}{d \ln \langle N_{ch} \rangle} \implies \langle p_T \rangle \propto \langle N_{ch} \rangle^{c_s^2}. \quad (2)$$

One caveat to this method has been identified in what is known as *centrality classification*, or the criterion used to classify collision events based on the degree of geometric overlap between the colliding nuclei. This criterion introduces a source of systematic uncertainty into the power-law relation due to the presence of a non-vanishing volume term into eq. (2). Utilizing a hydrodynamical collision model known as *Trajectum*, with centrality classification based on N_{ch} measured above a given momentum threshold p_T^{th} , I have identified a posi-

tive relation between c_s^2 and p_T^{th} (see FIG. 1a). I therefore propose that the power-law method of extracting c_s^2 should be compared to a novel, event-by-event response method [3] in order to identify the conditions in which geometric fluctuations are suppressed. This paper states that the *statistical noise* introduced to the linear response between event-mean p_T and N_{ch} is caused by event-by-event fluctuations in the initial state geometry. Through statistical analysis of the joint-distribution between the momentum and multiplicity fluctuations (shown in FIG. 1b), a method is proposed to mitigate the presence of these geometric fluctuations. My study takes advantage of this method to identify the precise p_T^{th} which is consistent with the mitigation of this noise.

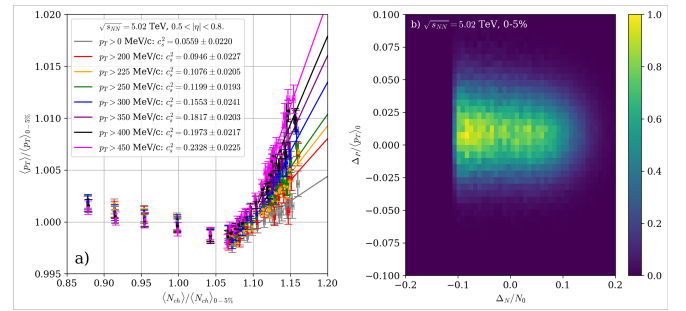


FIG. 1: N_{ch} - p_T correlation at collision energy $\sqrt{s_{NN}} = 5.02$ TeV in *Trajectum*. a): Power-law extraction of c_s^2 under different centrality classifier configurations, showing the p_T^{th} dependence within a given pseudorapidity η window. b): Distribution of the event-by-event multiplicity and momentum fluctuations normalized to the 0-5% centrality class.

The focus of my research is therefore to investigate the criteria under which the power-law method is consistent with the event-by-event response methodology. This analysis will be done using *Trajectum* and potentially other hydrodynamical collision models, as applied to the experimental conditions of the Relativistic Heavy Ion Collider and the Large Hadron Collider. The implication of this study is its potential to provide experimental criteria for a valid measurement of the thermodynamics of heavy-ion collisions. This is the key result that will be presented in a future *few-author paper*, which we aim to complete by the end of the summer semester. My contribution to this study will be particularly valuable once applied to different $\sqrt{s_{NN}}$, as it will inform the ongoing analyses within the STAR Collaboration.

[1] F. G. Gardim et al., Physics Letters B **856**, 138937 (2024).
[2] F. G. Gardim et al., Nature Physics **16**, 615 (2020).

[3] Y.-S. Mu et al., Physical Review Letters **135**, 162301 (2025).